



National University of Computer and Emerging Sciences



National University of Computer & Emerging Sciences, Peshawar

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Section: BSE-3A

Assignment: 2

Ramzan's Github: <https://github.com/ramzanbhutto/legacy-calc-2009.git>

Atiq's Github: <https://github.com/atiq-dehraj/legacy-calc-2009.git>



Loan EMI Calculator - Project Report

Project Overview:

The Loan EMI Calculator is a C++ application that calculates Equated Monthly Installments (EMI) for loans. This project demonstrates OOP principles, modern C++ build systems, comprehensive unit testing and automated documentation generation.

Technologies Used:

Technology	Purpose
C++	Core Programming language
CMake	Build system generator
Google Test	Unit testing framework
Doxygen	Documentation Generator
Git	Version Control

Key Features:

- Interactive Menu System
- Multiple Input Methods
- Accurate Calculations
- Comprehensive Testing
- Beautiful Documentation
- Modern Build System(CMake)



Bug Fixed and Refactoring

Old config.txt:

```
amount = 10000
interest = 12
years = 5
```

Wrong way

- Because the data member name is interest_rate not interest:

Fixed config.txt:

```
amount = 10000
interest_rate = 12
years = 5
```

- As it must matches the Code in Loan.cpp:

else if(key == "interest_rate") setInterestRate(value);

Impact: This bug prevented users from loading configuration files successfully.

Refactoring:

- Used std::unordered_map<std::string, long double> for data storage
- It is flexible and has key-based access.

```
class Loan{
private:
    std::unordered_map<std::string, ld> data; ///< stores loan data and results
public:
```

And some more like exception handling, input validation and key-value mapping.

Default Constructor with unordered_maps:

```
Loan::Loan(){
    data["amount"] = 0.0;
    data["interest_rate"] = 0.0;
    data["years"] = 0.0;
    data["monthly_rate"] = 0.0;
    data["months"] = 0.0;
    data["emi"] = 0.0;
    data["total_payment"] = 0.0;
    data["total_interest"] = 0.0;
}
```

Loading file with fstream and using stringstream to give input(string stream to int):

```
void Loan::loadConfig(const std::string& file){
    std::ifstream file1(file);
    if(!file1.is_open()){
        throw std::runtime_error("Could not open file: " + file);
    }

    std::string key, eq, line;
    ld value;
    while(std::getline(file1, line)){
        std::stringstream ss(line);
        ss>>key>>eq>>value;

        if(eq != "=") continue;
        if(key == "amount") setAmount(value);
        else if(key == "interest_rate") setInterestRate(value);
        else if(key == "years") setYears(value);
    }
}
```



Screenshot of Test Output

```
[ramzan@arch ~/legacy-calc-2009/build]$ test/unit_test
Running main() from /usr/src/debug/gtest/googletest-1.17.0/googletest/src/gtest_main.cc
[=====] Running 13 tests from 1 test suite.
[-----] Global test environment set-up.
[-----] 13 tests from LoanTest
[ RUN      ] LoanTest.DefaultConstructor
[ OK       ] LoanTest.DefaultConstructor (0 ms)
[ RUN      ] LoanTest.SettersGetters
[ OK       ] LoanTest.SettersGetters (0 ms)
[ RUN      ] LoanTest.isValidTrue
[ OK       ] LoanTest.isValidTrue (0 ms)
[ RUN      ] LoanTest.isValidFalseNegativeAmount
[ OK       ] LoanTest.isValidFalseNegativeAmount (0 ms)
[ RUN      ] LoanTest.isValidFalseNegativeYears
[ OK       ] LoanTest.isValidFalseNegativeYears (0 ms)
[ RUN      ] LoanTest.isValidTrueZeroInterestRate
[ OK       ] LoanTest.isValidTrueZeroInterestRate (0 ms)
[ RUN      ] LoanTest.CalculateEMIZeroInterestRate
[ OK       ] LoanTest.CalculateEMIZeroInterestRate (0 ms)
[ RUN      ] LoanTest.CalculateEMINonZeroInterestRate
[ OK       ] LoanTest.CalculateEMINonZeroInterestRate (0 ms)
[ RUN      ] LoanTest.CalculateEMIHighInterestRate
[ OK       ] LoanTest.CalculateEMIHighInterestRate (0 ms)
[ RUN      ] LoanTest.CalculateEMISmallLoanAmount
[ OK       ] LoanTest.CalculateEMISmallLoanAmount (0 ms)
[ RUN      ] LoanTest.LoadConfigSuccess
[ OK       ] LoanTest.LoadConfigSuccess (0 ms)
[ RUN      ] LoanTest.LoadConfigFileNotFound
[ OK       ] LoanTest.LoadConfigFileNotFound (0 ms)
[ RUN      ] LoanTest.TotalInterestCalculation
[ OK       ] LoanTest.TotalInterestCalculation (0 ms)
[-----] 13 tests from LoanTest (0 ms total)

[-----] Global test environment tear-down
[=====] 13 tests from 1 test suite ran. (0 ms total)
[ PASSED ] 13 tests.
```



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Git Commit History

Ramzan's History:

The screenshot shows the Git commit history for the user 'ramzanbhutto' on the repository 'legacy-calc-2009'. The interface is in dark mode. At the top, there's a navigation bar with links to Code, Pull requests, Actions, Projects, Wiki, Security, Insights, and Settings. Below this, the 'Commits' section is displayed for the 'dev' branch. The history is filtered by 'All users' and 'All time'. The commits are listed in chronological order, with the most recent at the top. Each commit entry includes a description, the user's name, the time ago, and a commit hash with a link to view the commit details.

Commit Description	User	Time Ago	Commit Hash
added FINAL Report	ramzanbhutto	1 minute ago	1a6c0f4
final check done	ramzanbhutto	22 minutes ago	a741095
edited .gitignore	ramzanbhutto	2 hours ago	7b6f4e4
Project Completed with Testing, Documentation and Building	ramzanbhutto	2 hours ago	c585dd5
completed unit(emi) test using gtest	ramzanbhutto	4 hours ago	5157431
Completed Files and added CMakeLists.txt	ramzanbhutto	5 hours ago	dc74955
Commits on Nov 25, 2025			
added README.md file	ramzanbhutto	yesterday	f7bf25b
Created folders, build structure and started	ramzanbhutto	yesterday	3f1010f

Atiq's History:

The screenshot shows the Git commit history for the user 'atiq-dehraj' on the repository 'legacy-calc-2009'. The interface is in dark mode. At the top, there's a navigation bar with links to Code, Pull requests, Actions, Projects, Wiki, Security, Insights, and Settings. Below this, the 'Commits' section is displayed for the 'dev' branch. The history is filtered by 'All users' and 'All time'. The commits are listed in chronological order, with the most recent at the top. Each commit entry includes a description, the user's name, the time ago, and a commit hash with a link to view the commit details.

Commit Description	User	Time Ago	Commit Hash
fixed gitignore file	atiq-dehraj	25 minutes ago	0eb64e5
done with documentation, testing and files	atiq-dehraj	27 minutes ago	0b5dfff
build structure	atiq-dehraj	3 hours ago	2014108



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[ramzan@arch ~/legacy-calc-2009]\$ git log --oneline --graph --decorate --all

```
* 7bef4e4 (HEAD -> dev, origin/dev) edited .gitignore
* c585dd5 Project Completed with Testing, Documentation and Building
* 5157431 completed unit(emi) test using gtest
* dc74955 Completed Files and added CMakeLists.txt
* f7bf25b added README.md file
* 3f1010f Created folders, build structure and started
* 347e865 (origin/master, origin/HEAD, master) Added compilation information
* dff0427 Nicer formatting on Readme and added better app help text
* 71f0456 Added readme and additional app help text
* c3e0801 Added a Qt Gui and scons compilation. The makefile was generated by Qt.
* be10569 Changes to use libCmdLineParser
* 12ee333 Removed compileCommands.txt
* 4d09e8b Made methods inline and removed some compiler errors
* d99e446 First version
```



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Pull Request:

The screenshot shows a GitHub Pull Request (PR) for the repository 'bradyallenjohnson / LoanCalculatorCpp'. The PR title is 'Fixed input validation from user and file and added CMake configs, Doxygen documentation support, maps, google unit tests, and a beautiful README file #6'. The PR is open and shows 7 commits from 'ramzanbhutto:dev' to 'bradyallenjohnson:master'. The PR description includes a table of contents for the 'Loan EMI Calculator' project, which is a comprehensive C++ loan calculator. The table of contents lists: Overview, Features, Project Structure, What I Built & Configured, Prerequisites, Building the Project, Running the Application, Running Tests, and Generated Documentation. The PR also shows a sidebar with 'Reviews' (No reviews), 'Assignees' (No one assigned), 'Labels' (None yet), 'Projects' (None yet), 'Milestone' (No milestone), and 'Development' (Successfully merging this pull request may close these issues).

Fixed input validation from user and file and added CMake configs, Doxygen documentation support, maps, google unit tests, and a beautiful README file #6

ramzanbhutto wants to merge 7 commits into bradyallenjohnson:master from ramzanbhutto:dev

Conversation 0 Commits 7 Checks 0 Files changed 239 +12,570 -60

ramzanbhutto commented yesterday • edited

Loan EMI Calculator

A comprehensive C++ loan calculator that computes Equated Monthly Installments (EMI) with a clean object-oriented design, automated testing, and full documentation.

Note: The QT_Legacy folder is not part of this project.

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Overview

ramzan

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